

24*24 Super-small Development Board

Specification and Instruction

Content : Specification and Introduction manual of SIMCOM 24*24
packaging super-small development board

Date: 2023-07-03

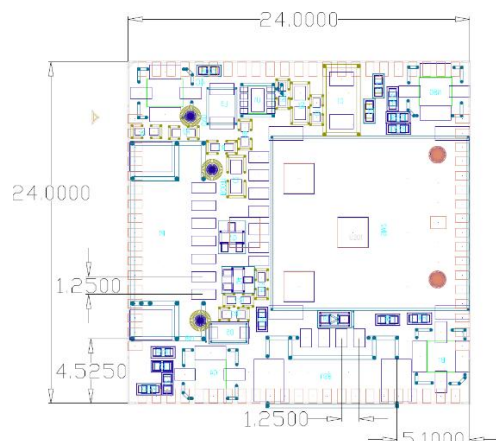
Creator: Jack

Dimension of Super-small Development Board

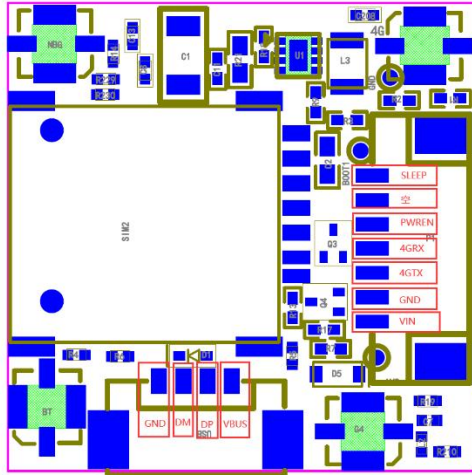
1.Appearance and Fittings



2.Dimension of the HW



3.Declaration of the Pin order



Dimension of the Pin P1=1.25mm

USB Pin

1: GND Grounding

2: DM Connect USB DM

3: DP Connect USB DP

5: VBUS Connect USB VBUS---(P.S. VBUS is only used for connecting USB,
can not be used for supplying electricity for PCBA)

P1 Pin

1.VIN: input power , Range of the Voltage of the Power 1.2V-4.2V (Pay
attention!)

2.GND: Grounding

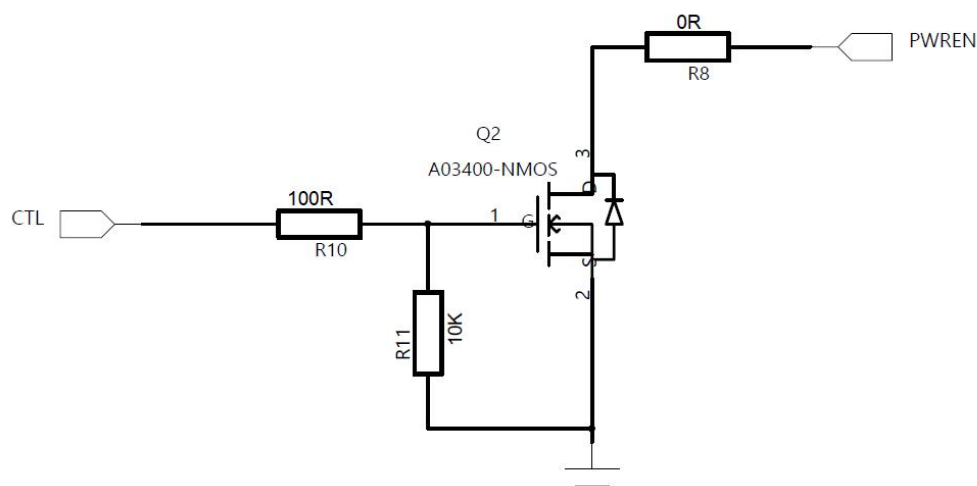
3.4GTX: Connecting the serial port RX of the MCU

4.4GRX: Connecting the serial port TX of the MCU

5.PWREN: Control the power of the PCBA, the main board need to add transistor to control electrical level and the circuit (lower or turn off the power of the PCBA)

P.S.

The following graph is the control design for your reference.

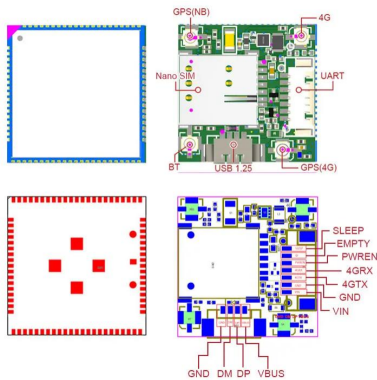


6.Empty Pin

7.SLEEP: Control Sleep, use this Pin to activate the modem after sending sleep command to 4G modem, the common high electrical level is 1.8V, lower the GNS can help wake up.

Definition of the HW interfaces

Specification of the internal components



Adaptive modem: SIMCOM(A7670X, SIM7670X, NB/SIM7070X, SIM7000X)

IPEX Antenna: First generation RF base

Adaptive antenna: built-in FPC/PCB/External antenna

Interfaces: USB/ Serial Port

SIM card slot: Nano SIM 7 PIN PUSH-PUSH type card slot

Accessories: 7P DuPont thread, 4P to Type-C Cable

Method of Testing

When testing, if you use serial port please connect it with 7P Dupont thread, use TTL serial port adaptive board to connect with VIN and GND to provide modem with 3.3V voltage. The serial port connect with TX and RX and then can work. You can also use USB port to test, but please pay attention the USB port could not be used to supply electricity, when you use USB port to test you need to add an external power to supply electricity. The voltage range of LTE modem is 1.2V~4.2V, we recommend you to use 3.3V. The

voltage range of NB-IoT modem is 0.8V~4.2V. You can choose by your need.

P.S.

This specification is the datasheet of the 24X24 Super-small Board, if you want the detailed datasheet of the main modem used on the board, please look up the datasheet according to its part number.